Supervised learning

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PART - A : Conceptual Understanding (Theory)

Q1 What are Supervised Learning Algorithms?

→ Supervised learning algorithms learns pattern from labeled data, where both input features and the correct output are known.
The model learns a mapping function between inputs and outputs and can predict results for new unseen data.

Q.2. Difference between Regression Algorithms vs Classification Algorithms.

→

Regression	Classification
> Predict continuous values.	> Predict categories/ classes
> Ex:- house price	> Ex :- spam or not spam
> Output is numeric	> Output is discrete labels.

Q.3. Explain simple linear Regression.

→ Simple linear Regression models the relationship between one independent variable and one dependent variable.

$$y = \beta_0 + \beta_1 x + \epsilon$$

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> y = target variable
x = feature
β_0 = intercept
β_1 - slope
ϵ = error term

Q.4. List and explain the assumption of linear Regression.

→ 1. Linearity - The relationship between predictors (x) and the outcome (Y) is straight line.

2. Independence - observation are independent, not influencing one another.

3. Homoscedasticity - Residual (errors) have constant variance across all level of x.

4. Normality - Residual are normally distributed.

Q.5. What is bias - Variance Trade-Off?

→ The bias-variance tradeoff is the balance between a model's simplicity (bias) and complex (variance) to minimize total prediction error.

- Bias (underfitting): Error from overly simple assumption (e.g., linear model for curved data). High bias means the model ignores relevent relations.

- variance (overfitting) - Error from too much complexity (e.g., a high-degree polynomial). High variance means the model is too sensitive to small fluctuation in training data.

- Trade Off : As bias decreases (model becomes more complex), variance increase, and vice versa, The goal is to find the "Goldilocks zone" where both are low.

Q.6. Explain Overfitting and underfitting with example.

→ underfitting

- Model too simple
- high bias
- Poor performance on Train & Test.

Overfitting

- model too complex
- High variance
- very good on train but bad on test